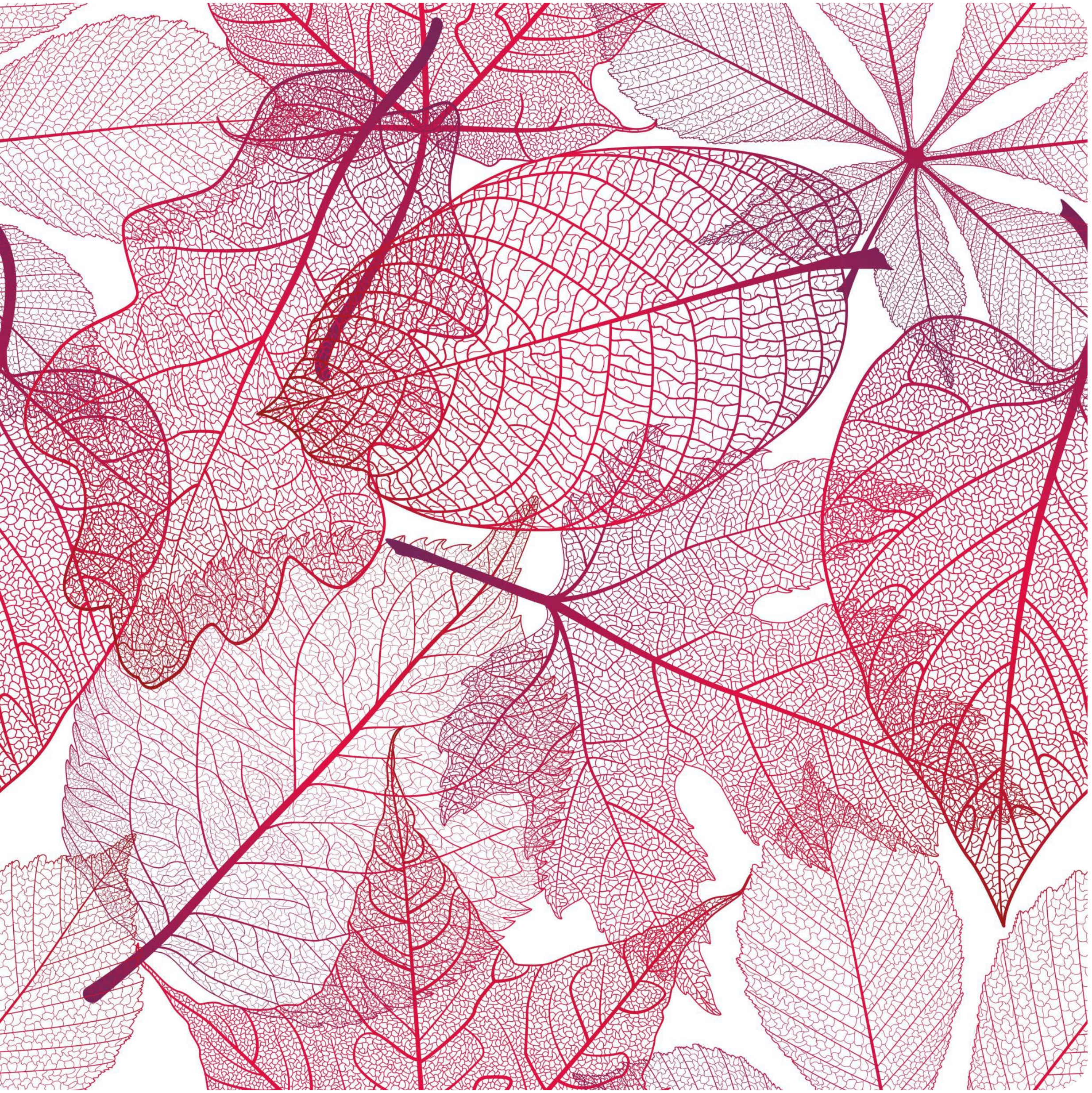


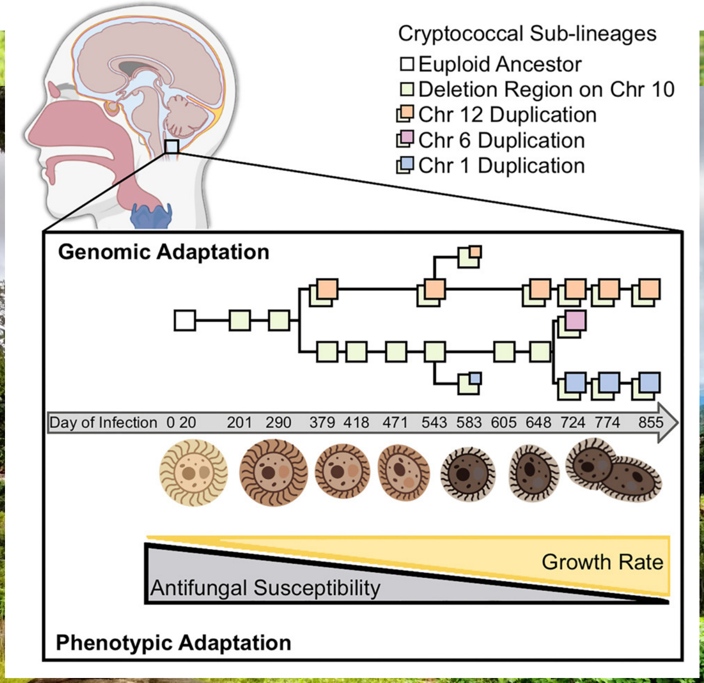
BIO 724D Data Lunch

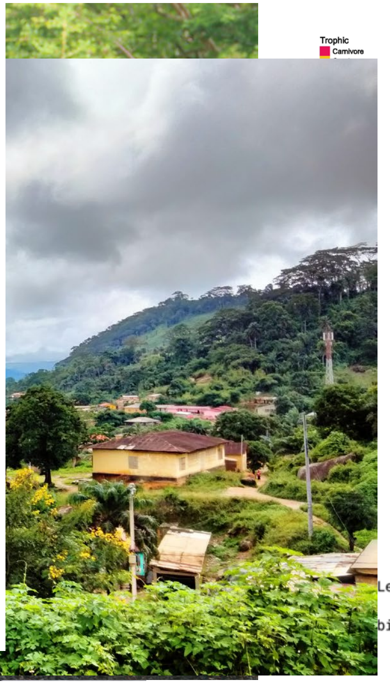
Kayla Wilhoit

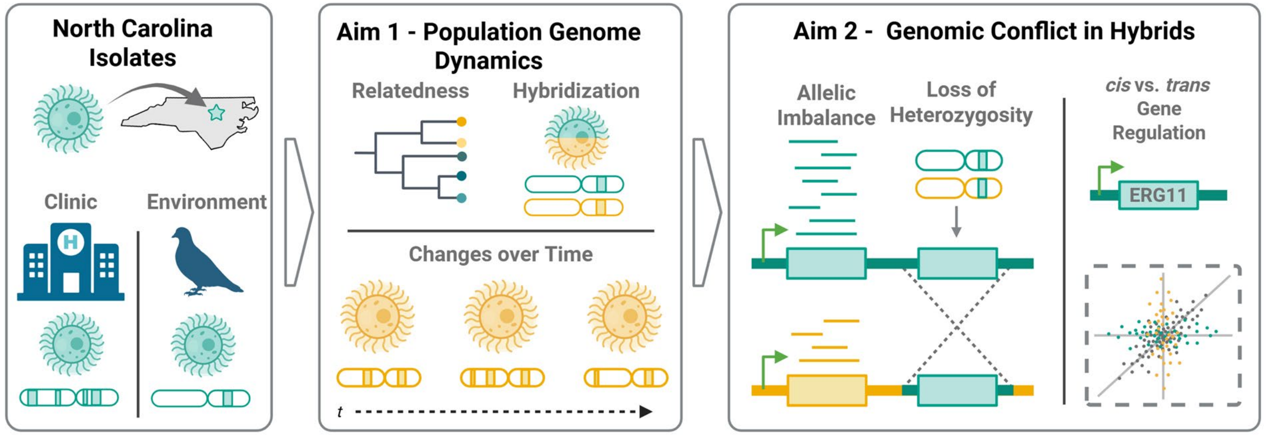
University Program in Genetics and Genomics

Magwene Lab

Original XY System







- Dallas, Texas
- Interest in biology and genetics
- No computational skills :(

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neopets®

MINECRAFT

- Dallas, Texas
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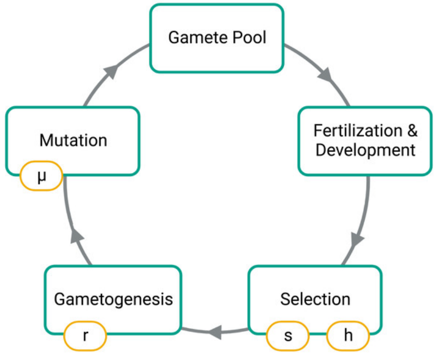
- Texas A&M University
  - Biomedical Sciences Major
  - Genetics Minor
- R, RStudio, experimental design, basic statistics
- Population genetics simulations in R



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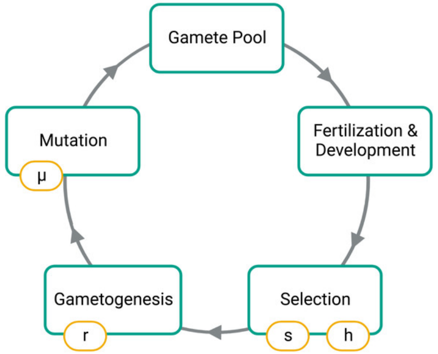


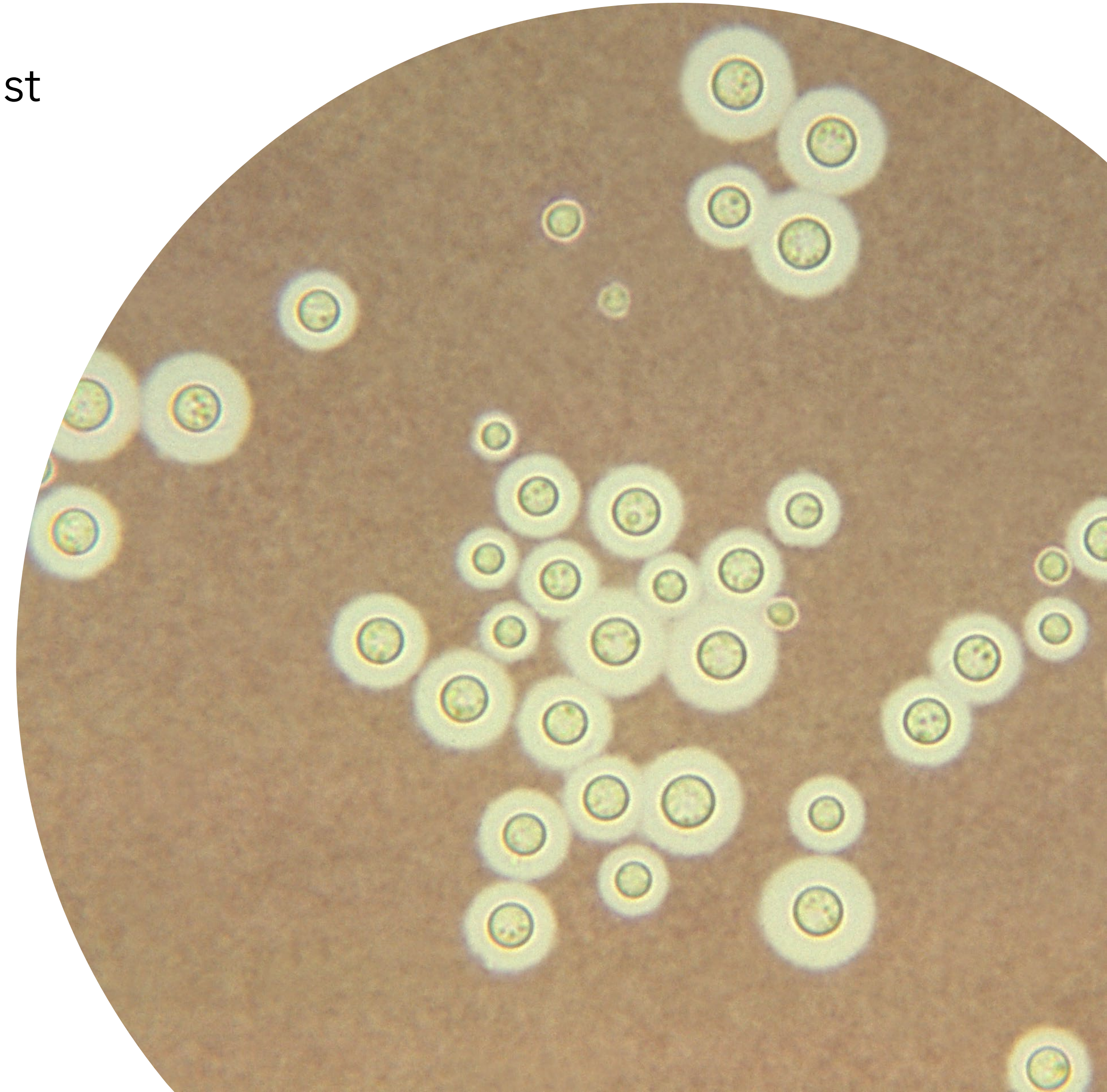
Duke

University Program *in* Genetics *and* Genomics

- Duke UPGG Rotations
  - Wray lab – Intro to Unix, command line, genomics tools
  - Magwene lab – More Unix, python, snakemake,
  - Lowe lab – More snakemake, slurm, bash scripting
- Joined Magwene lab in Spring 2024
- Took BIO740 in Spring 2024

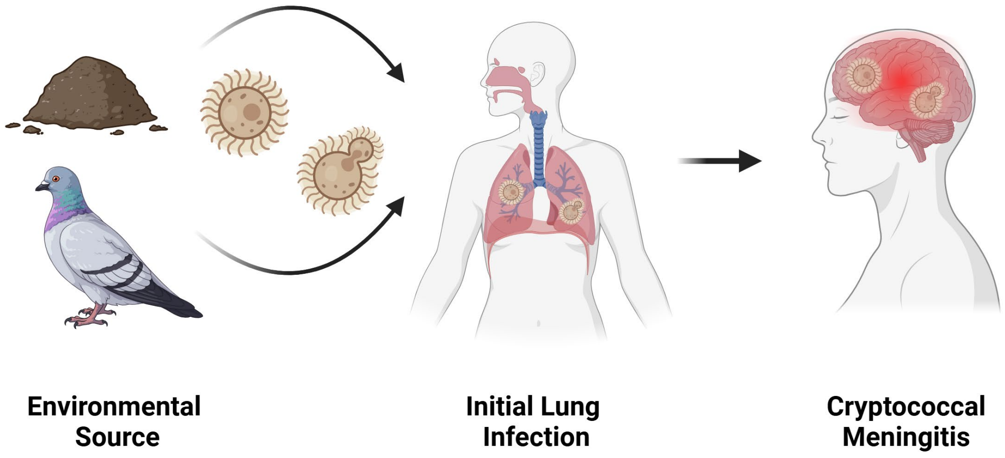
Research Interests

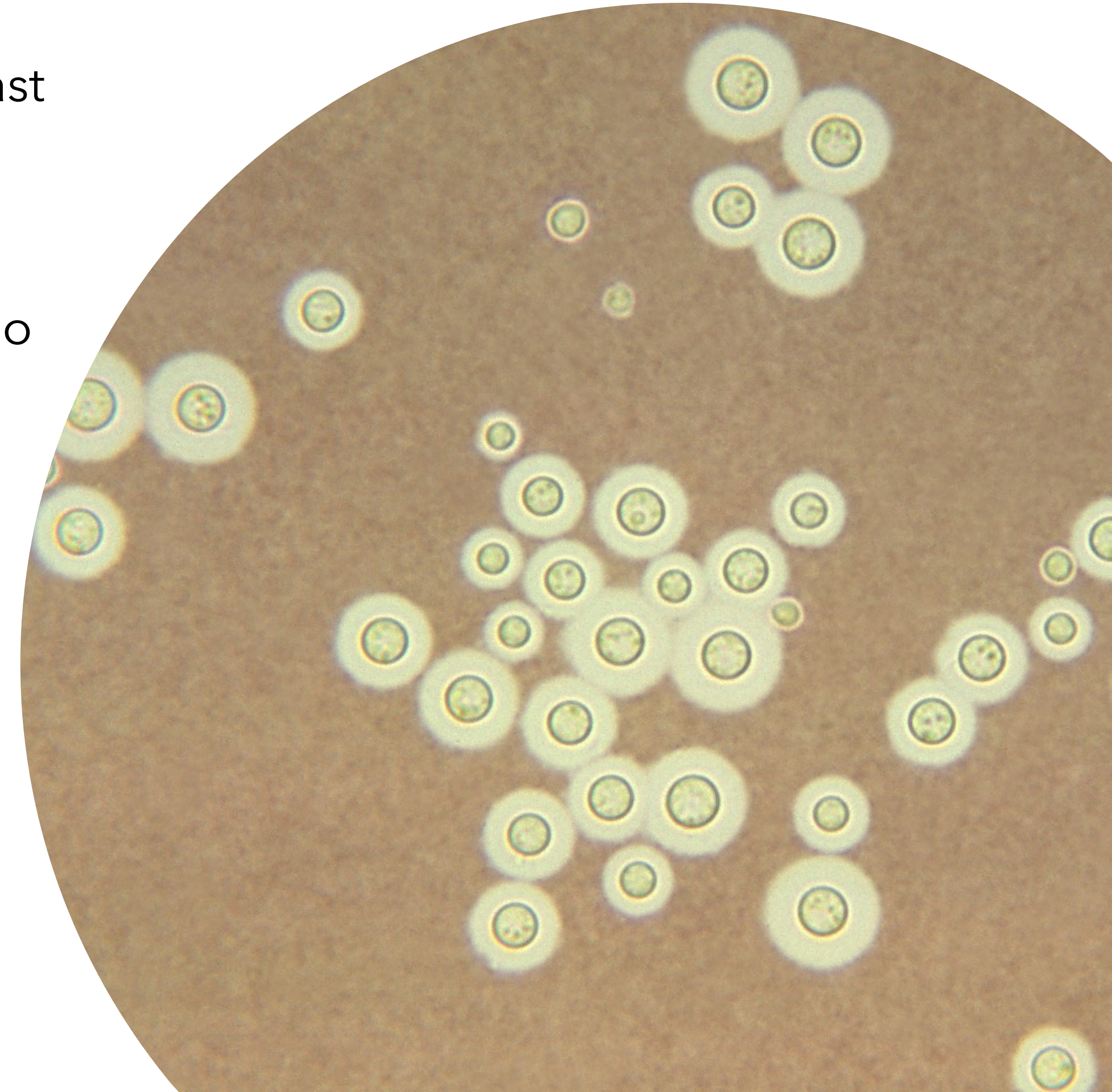
- *Cryptococcus neoformans* is a pathogenic fungal yeast that causes over 180,000 deaths annually
  - Primarily affects immunocompromised patients
  - Current antifungal treatments are often ineffective



Research Interests

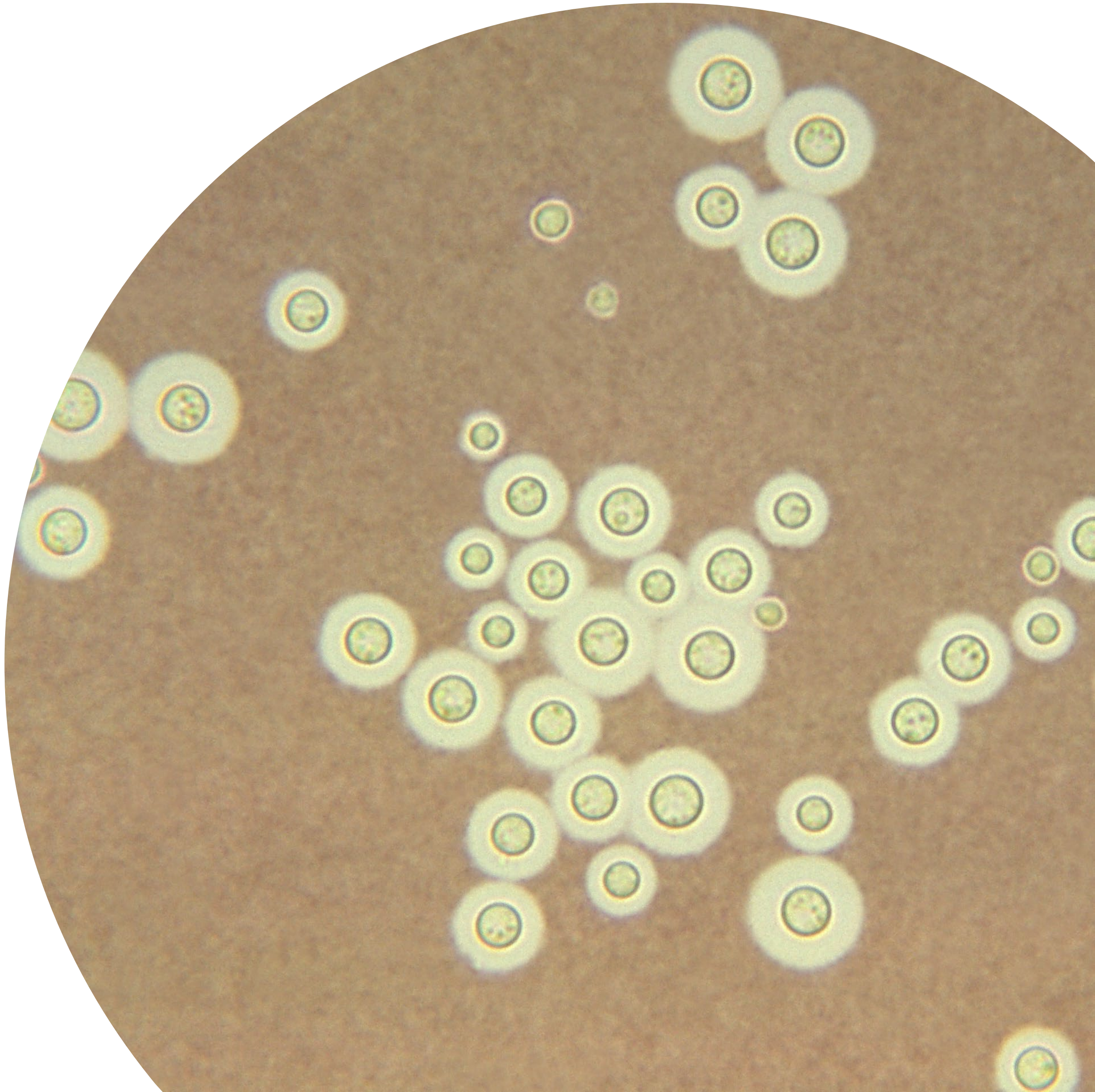
- *Cryptococcus neoformans* is a pathogenic fungal yeast that causes over <u>180,000 deaths annually</u>
  - Primarily affects immunocompromised patients
  - Current antifungal treatments are often ineffective
- Environmental sources include soil and pigeon guano
- High genetic plasticity and variation
  - 14 chromosomes, usually haploid
  - ~19 Mb genome



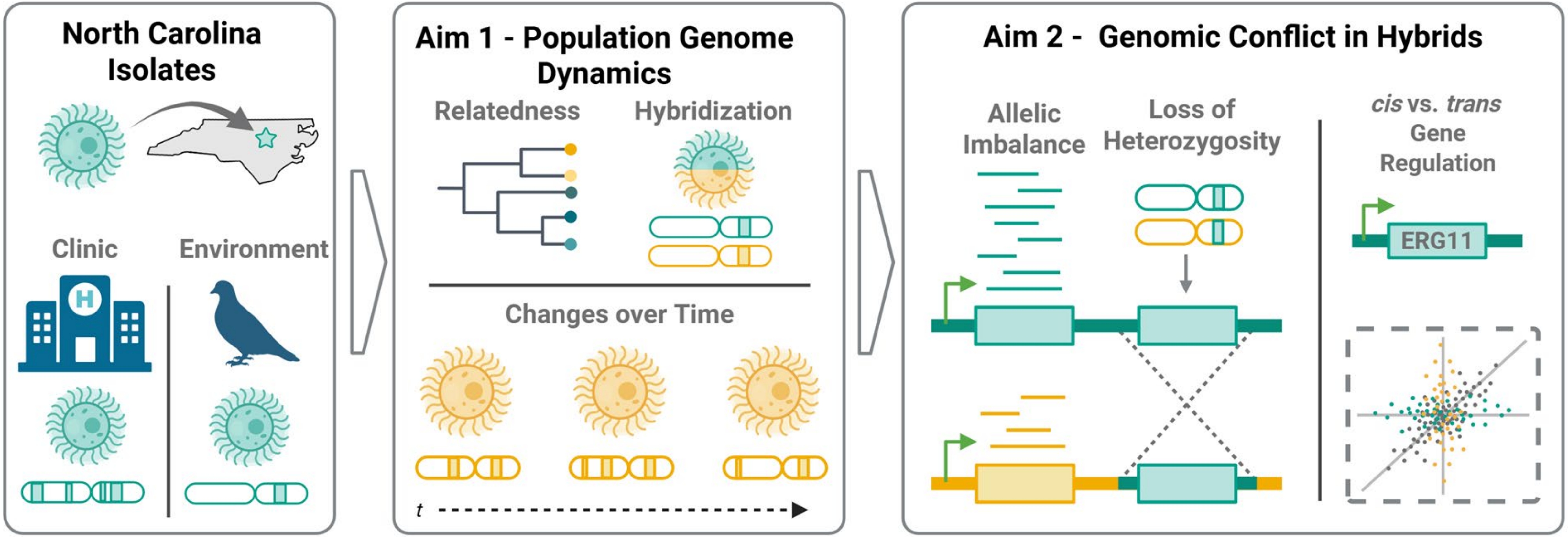


Research Interests

- Evolution of and selection on fungal hybrids in *Cryptococcus neoformans*
- Population dynamics of clinical and environmental *Cryptococcus* in North Carolina
- Mechanisms of genome regulation in diploid and hybrid *Cryptococcus*



Example 1 - Project Background

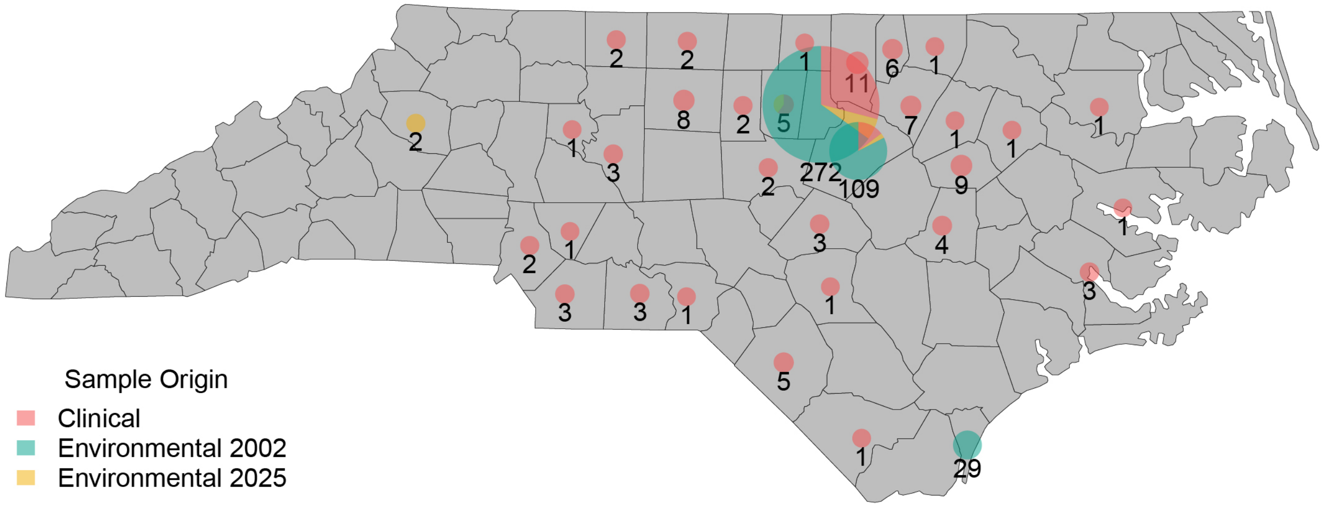


North Carolina Samples

- **349** sequenced clinical isolates
  - + more from Duke Hospital
  - **40** diploid hybrids
- **291** sequenced environmental isolates
  - **53** diploid hybrids
- **Are there genomic difference between clinical and environmental isolates?**

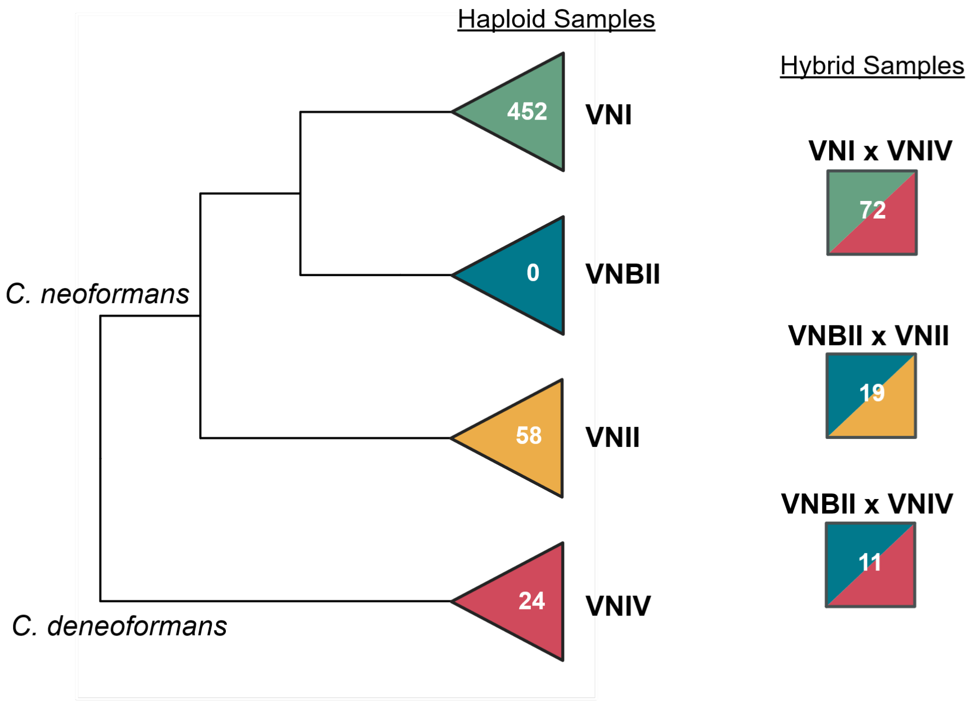
Anastasia Litvintseva, PhD

Marhiah Montoya, PhD



Computational Problems

- Lots of big genome files!
  - Naming conventions
  - Location consistency
  - Subsetting groups of samples for efficient analysis
- Dealing with hybrid genomes
- Integrating multiple genomics tools into one pipeline
- Producing useful intermediate files
- Visualizing results quickly



Favorite Tools

Visual Studio Code
<3

GitHub Copilot

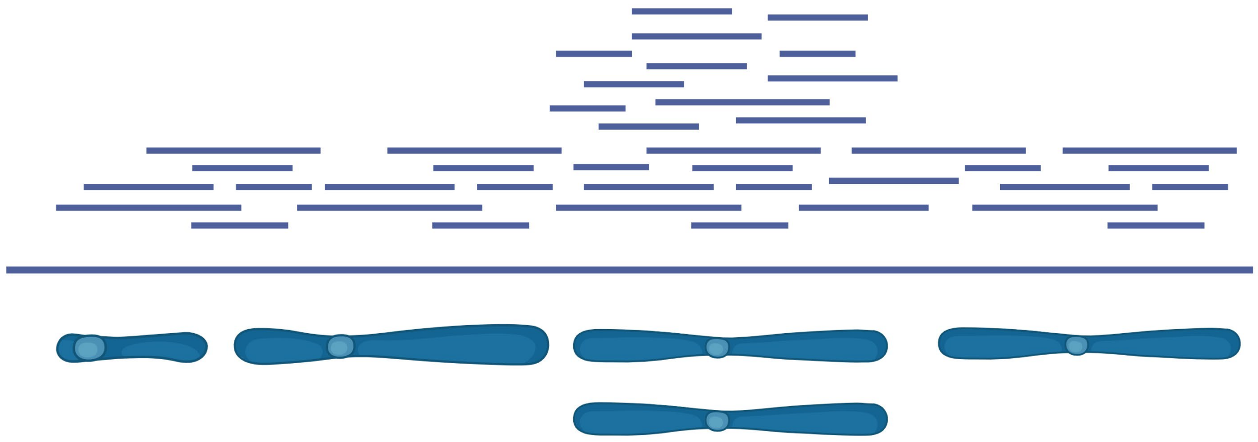
Anaconda

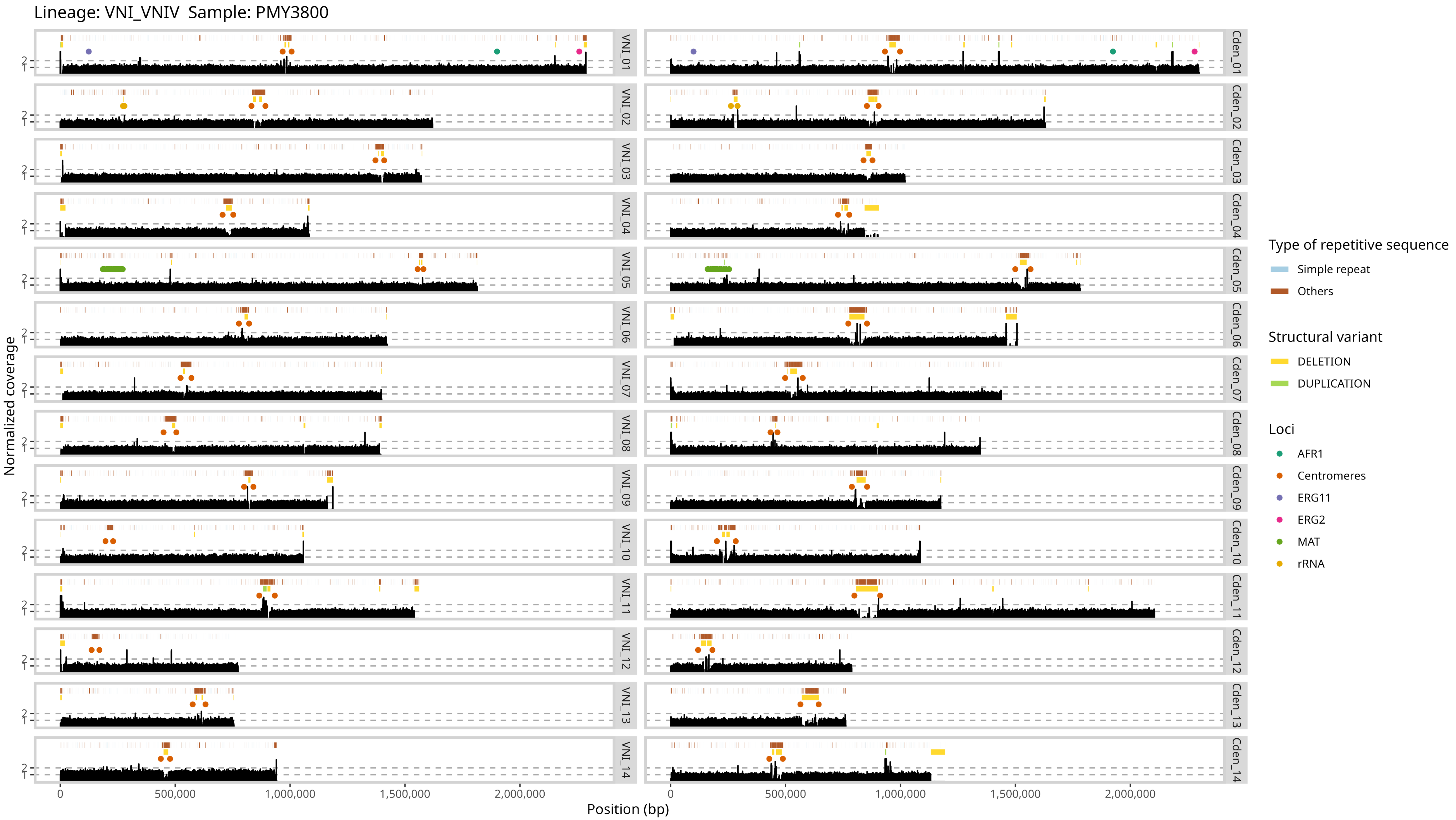
Windows Subsystem for Linux (Ubuntu)

Snakemake

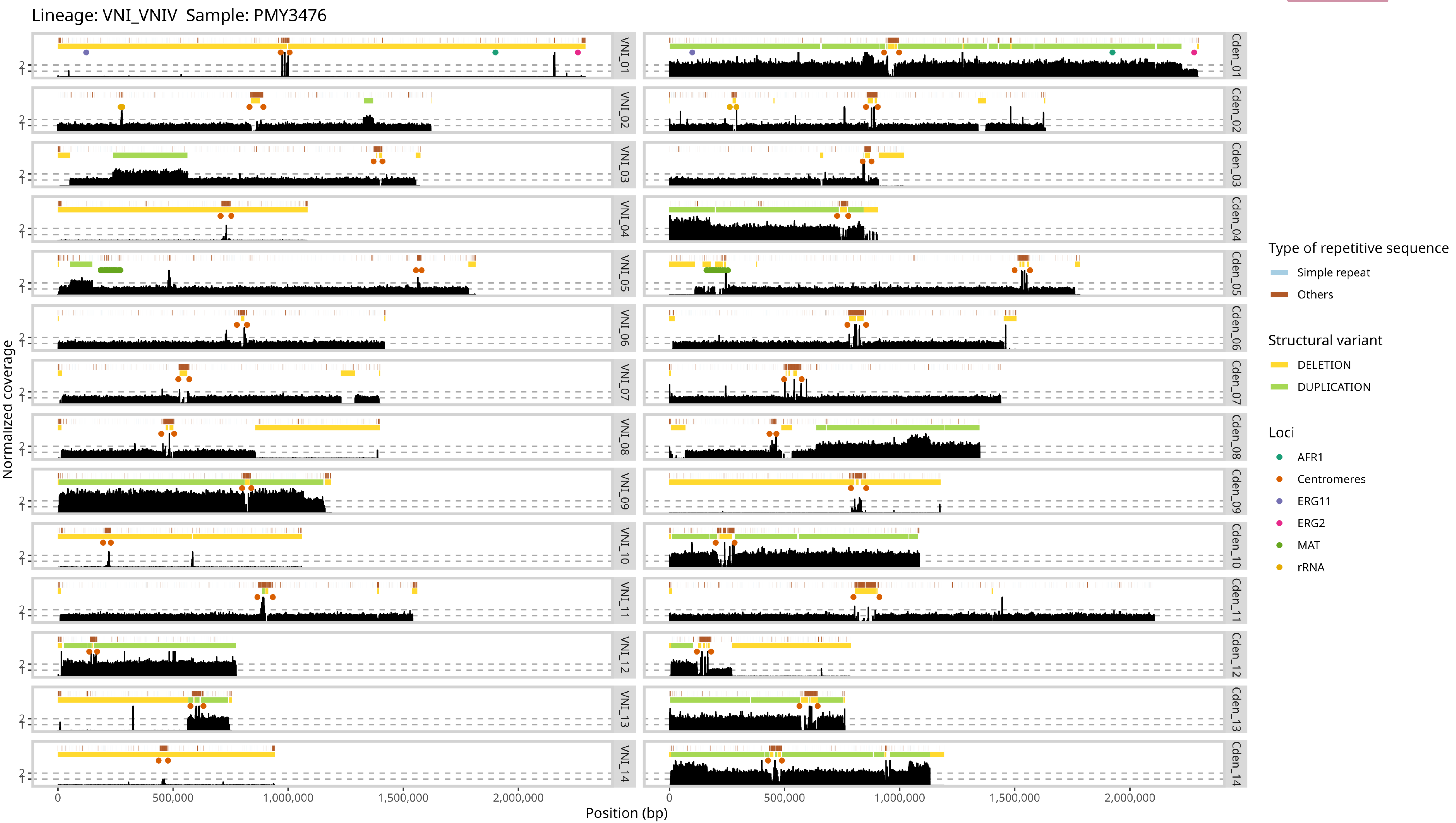
Read Depth Plots

- Align Illumina short-reads to concatenated reference genome
- Normalize reads to background
- Quantify read depth across the genome





Lineage: VNI_VNIV Sample: PMY3800
Normalized coverage
Position (bp)
0
500,000
1,000,000
1,500,000
2,000,000
VNI_01
VNI_02
VNI_03
VNI_04
VNI_05
VNI_06
VNI_07
VNI_08
VNI_09
VNI_10
VNI_11
VNI_12
VNI_13
VNI_14
Cden_01
Cden_02
Cden_03
Cden_04
Cden_05
Cden_06
Cden_07
Cden_08
Cden_09
Cden_10
Cden_11
Cden_12
Cden_13
Cden_14
Type of repetitive sequence
Simple repeat
Others
Structural variant
DELETION
DUPLICATION
Loci
AFR1
Centromeres
ERG11
ERG2
MAT
rRNA

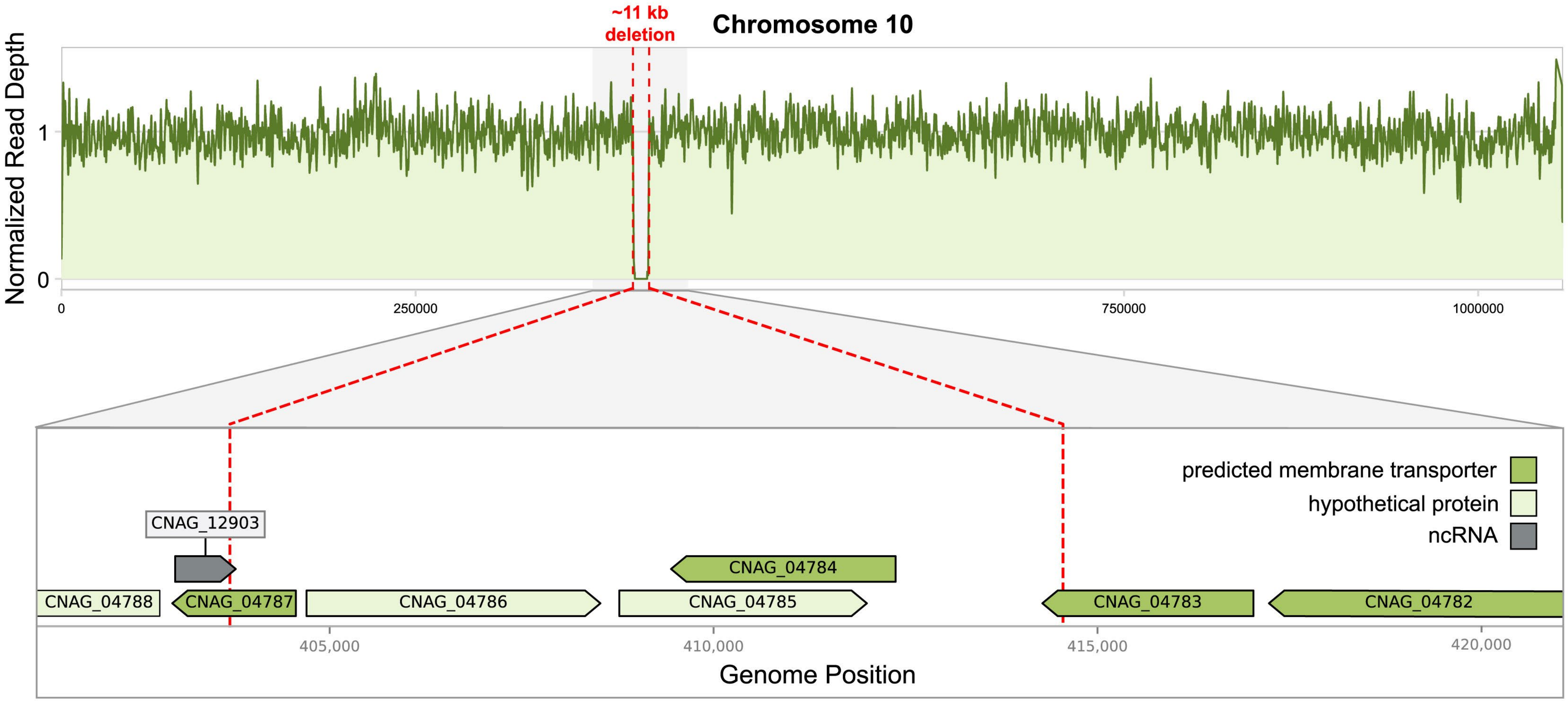


Lineage: VNI_VNIV Sample: PMY3476
Normalized coverage
Position (bp)
VNI_01
VNI_02
VNI_03
VNI_04
VNI_05
VNI_06
VNI_07
VNI_08
VNI_09
VNI_10
VNI_11
VNI_12
VNI_13
VNI_14
Cden_01
Cden_02
Cden_03
Cden_04
Cden_05
Cden_06
Cden_07
Cden_08
Cden_09
Cden_10
Cden_11
Cden_12
Cden_13
Cden_14
0
500,000
1,000,000
1,500,000
2,000,000
Type of repetitive sequence
Simple repeat
Others
Structural variant
DELETION
DUPLICATION
Loci
AFR1
Centromeres
ERG11
ERG2
MAT
rRNA

Read depth with **ggplot2** line plot + fill in R

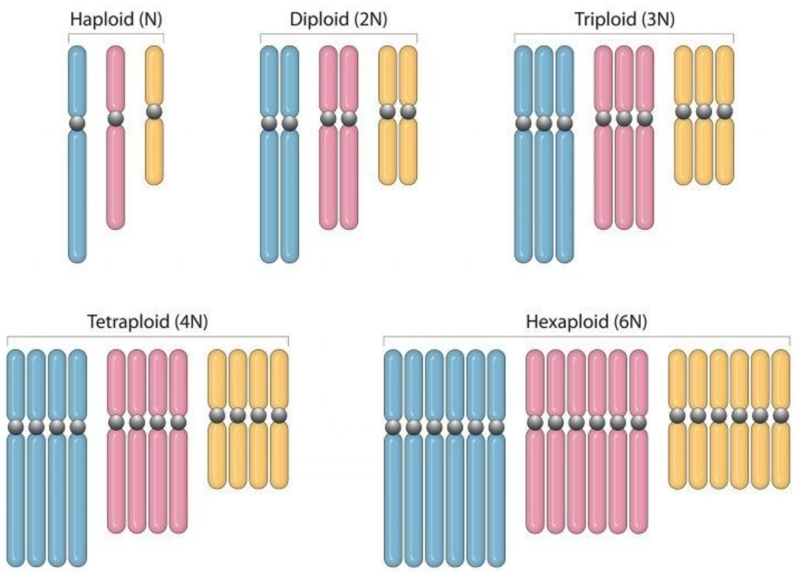
SVG plots combined and annotated in **Adobe Illustrator**

Gene track with **Gviz** package in R



Example 2 – Project Background

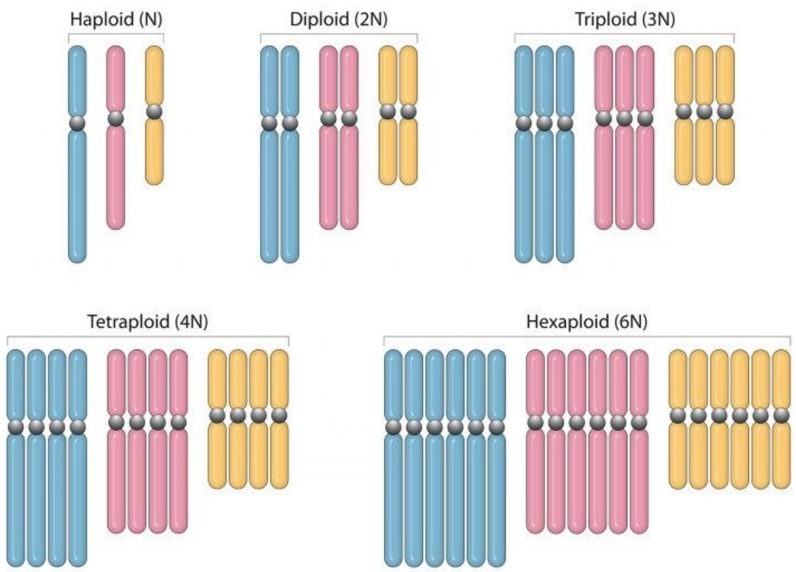
- *Cryptococcus neoformans* is typically haploid and clonal
  - Traditional crossing-over recombination and meiosis only occurs during specific mating conditions

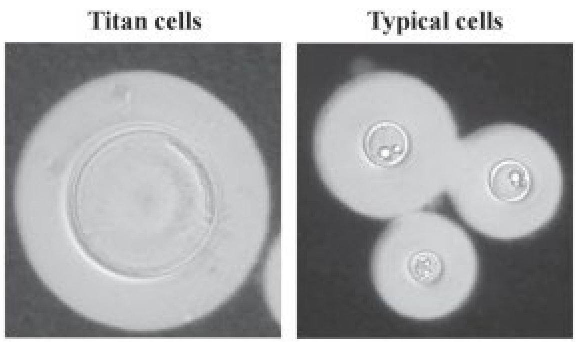


Example 2 – Project Background

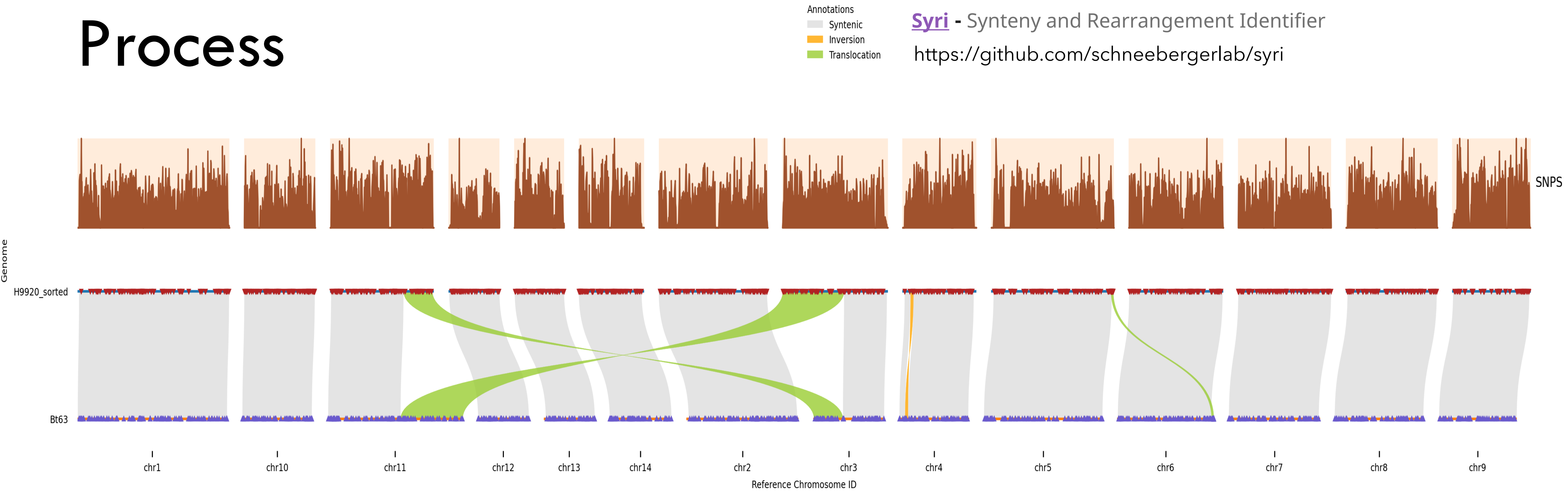
- *Cryptococcus neoformans* is typically haploid and clonal
  - Traditional crossing-over recombination and meiosis only occurs during specific mating conditions
- Some C. *neoformans* cells increase in ploidy and become massive under stress (titanization)
  - These polyploid titan cells can shed off diploid daughter cells that look normal

- **Is recombination occurring in titan cells?**





Process



Syri - Synteny and Rearrangement Identifier

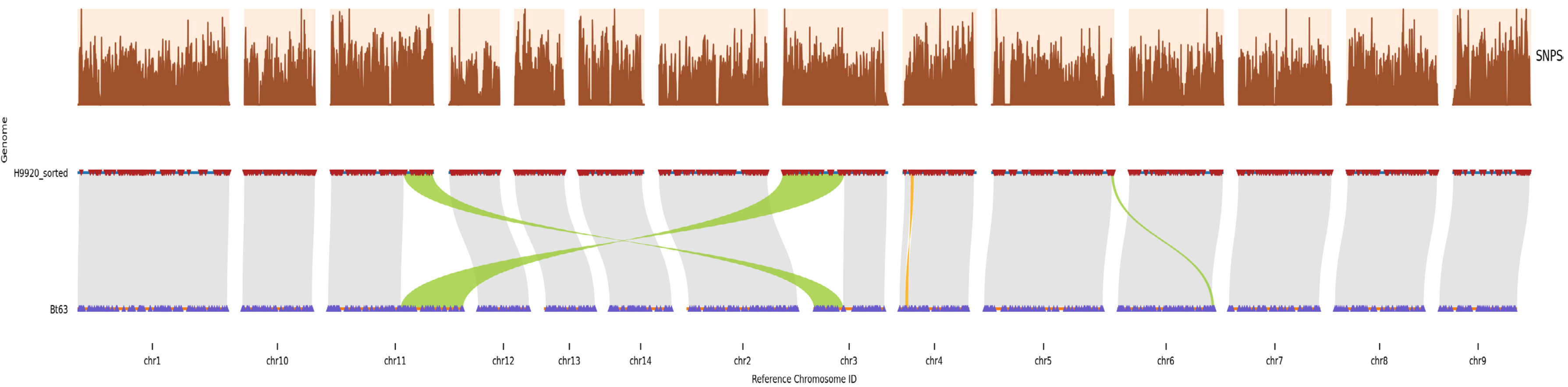
https://github.com/schneebergerlab/syri

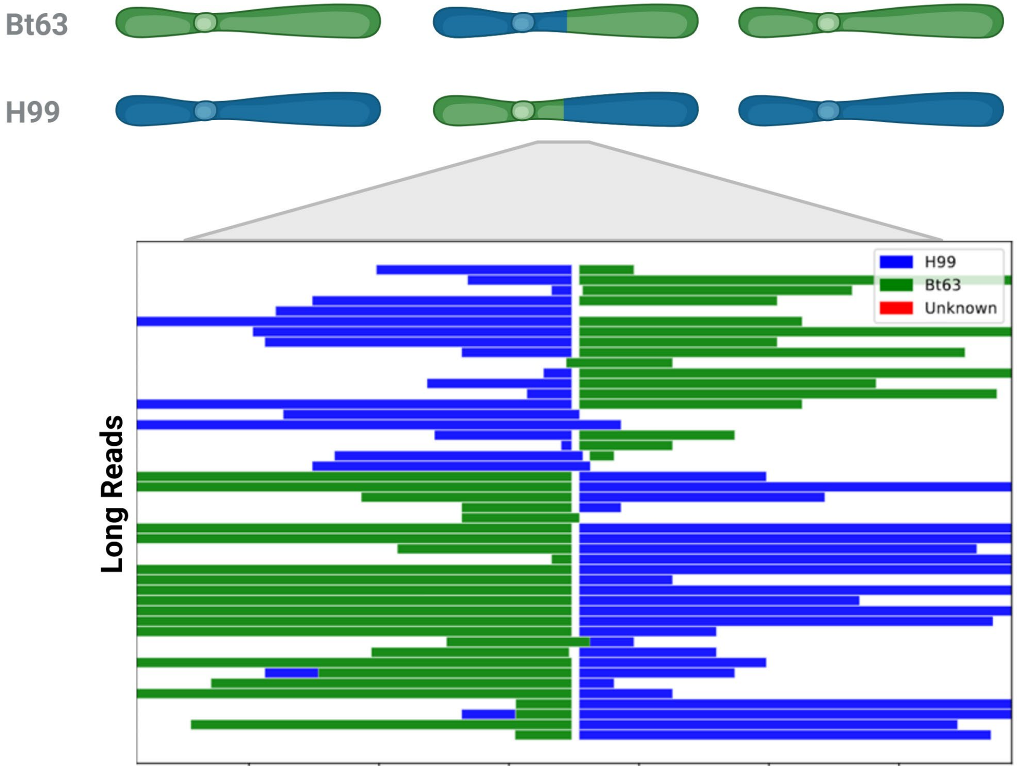
- Leverage SNPs distributed across the genome to identify origin of individual reads

Process

[Syri](https://github.com/schneebergerlab/syri) - Synteny and Rearrangement Identifier

https://github.com/schneebergerlab/syri





- Leverage SNPs distributed across the genome to identify origin of individual reads

Process

Align Progeny Reads

- Align long reads to H99 reference genome
- Gives location of progeny reads relative to reference

Intersect with Variants

- Find the list of variants between H99 and Bt63 that should be located within each read

chrom	start	end	name	f_chrom	f_start	f_end	h99	bt	f_type
chr1	5367	27883	S1_5846	chr1	25740	25740	T	C	SNP
chr1	5367	27883	S1_5846	chr1	25752	25752	C	T	SNP
chr1	5367	27883	S1_5846	chr1	25760	25760	A	C	SNP
chr1	5367	27883	S1_5846	chr1	25817	25817	G	A	SNP
chr1	5367	27883	S1_5846	chr1	25818	25818	A	T	SNP
chr1	5367	27883	S1_5846	chr1	25823	25823	A	G	SNP
chr1	5367	27883	S1_5846	chr1	25824	25824	T	C	SNP
chr1	5367	27883	S1_5846	chr1	25897	25897	C	T	SNP
chr1	5367	27883	S1_5846	chr1	26028	26028	T	C	SNP

Call Parent for each Variant

- Determine whether the base on the read at the variant position matches H99 or Bt63

```
S5_1355,chr5,91545,c,c,c,t,H99
S5_1355,chr5,91722,t,t,t,c,H99
S5_1355,chr5,92104,a,a,a,g,H99
S5_1355,chr5,92284,t,t,t,c,H99
S5_1355,chr5,92314,a,a,a,c,H99
S5_1355,chr5,92362,t,t,t,c,H99
S5_1355,chr5,92578,a,a,a,g,H99
S5_1355,chr5,92582,c,c,c,t,H99
S5_1355,chr5,92603,a,a,a,t,H99
S19_7453,chr5,90890,t,t,a,t,Bt63
S19_7453,chr5,90905,a,a,g,a,Bt63
S19_7453,chr5,90975,g,g,c,g,Bt63
S19_7453,chr5,91071,t,t,c,t,Bt63
S19_7453,chr5,91134,c,c,t,c,Bt63
S19_7453,chr5,91536,t,t,c,t,Bt63
S19_7453,chr5,91545,t,t,c,t,Bt63
S19_7453,chr5,91722,c,c,t,c,Bt63
```

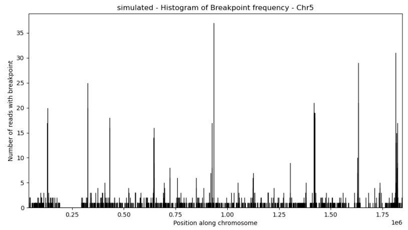
Create Intervals

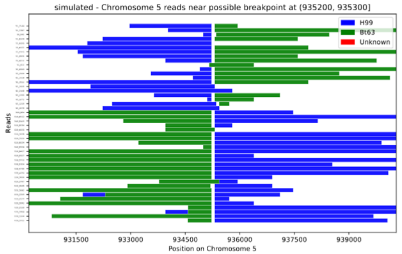
- Combine intervals where the parent is the same within each read
- Remove errors

```
S19_100,1373016,1374793,H99,19
S19_1000,371601,371995,Bt63,14
S19_1001,1047266,1047680,H99,16
S19_1002,1541436,1544964,H99,81
S19_1003,355656,355795,H99,4
S19_1003,355803,356689,Bt63,18
S19_1005,1755999,1759583,H99,64
S19_1006,1265383,1265610,H99,5
S19_1007,586698,588347,Bt63,24
S19_1008,1360815,1364855,H99,123
S19_1009,717728,721600,Bt63,103
S19_101,1226526,1227267,H99,8
S19_1010,148636,153625,Bt63,112
S19_1011,809400,811972,Bt63,56
S19_1012,895682,907482,Bt63,222
```

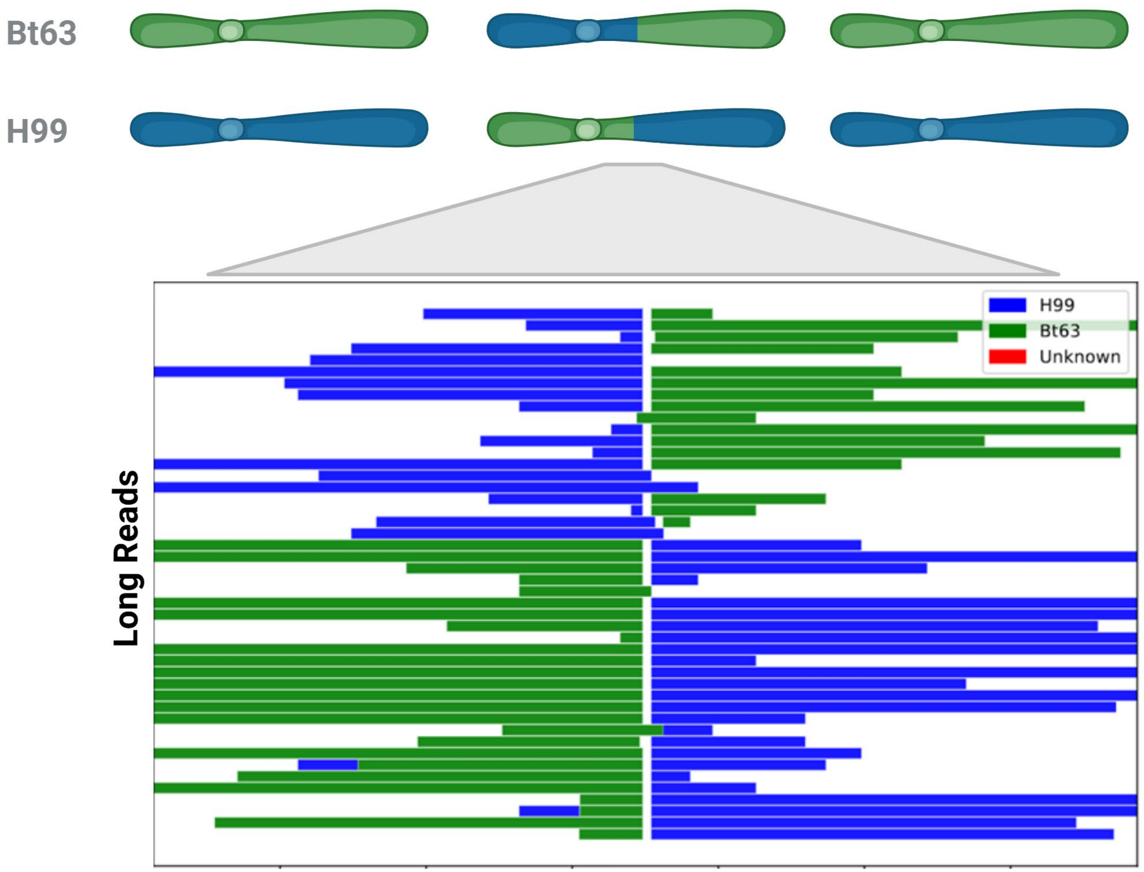
Find Breakpoints

- Find regions where multiple reads change from one parent to the other
- Plot output

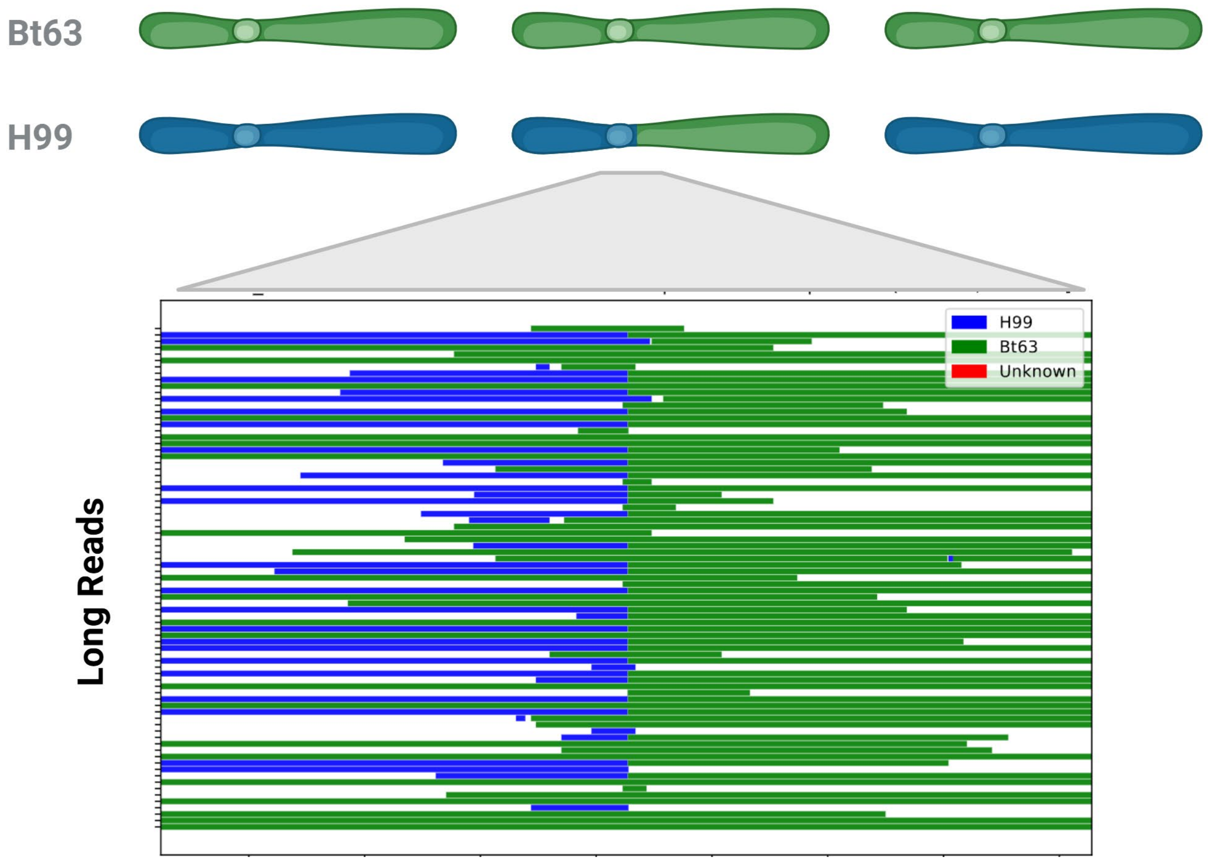




Simulated Data
(Crossing Over)



Actual Data
(Loss of Heterozygosity)



Genomics Software and Packages

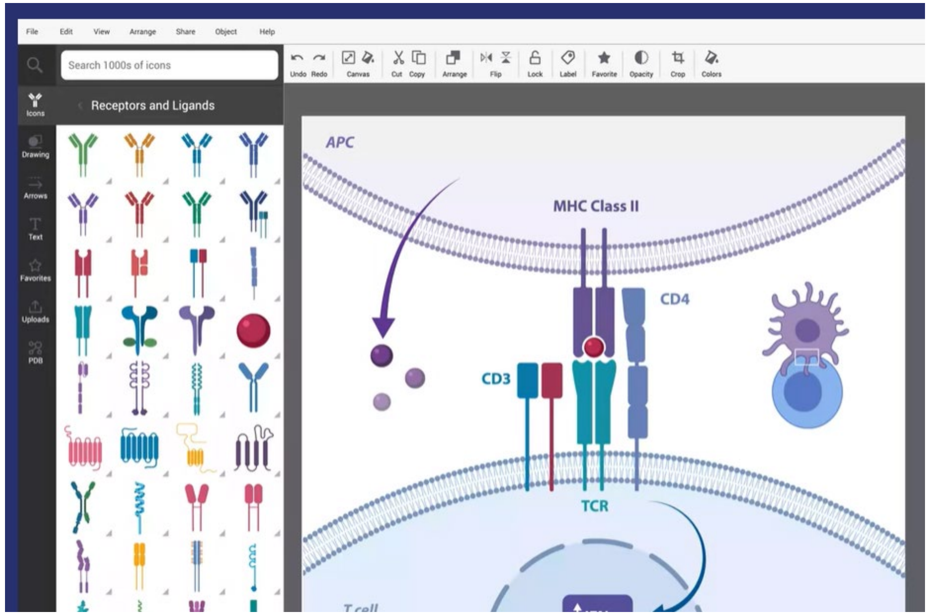
- Genome tools (command line)
  - **minimap2**
  - **mosdepth**
  - **seqkit**
  - **lra**
  - **samtools**
  - **bedtools**
  - **wfmash**
  - **syri**
  - **snippy/snpeff**
- Python packages
  - **pandas**
  - **pysam**
  - **pybedtools**
  - **multiprocessing**
- R packages
  - **tidyverse**
  - **ggplot2 <3**
  - **patchwork**

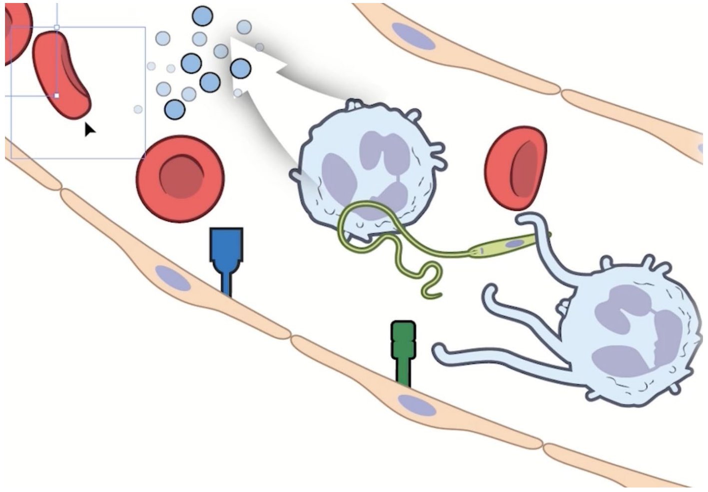
Useful VSCode Extensions

- Remote Development
  - **Remote – SSH: Editing Configuration Files**
  - **Remote Explorer**
  - **Remote – Tunnels**
    - For WSL
  - **Remote Repositories**
    - Copy/download files directly from repositories without cloning!
- Language-specific
  - **Jupyter (+ cell tags, renderers, slide show)**
  - **Python (+ Pylance, Debugger)**
  - **R (+ Debugger)**
  - **Snakemake Language**
- Quality of Life
  - **Rainbow CSV**
  - **Edit CSV**
    - Excel-like sheets interface
  - **Data Wrangler**
    - Powerful python-based data frame manipulation interface!
  - **SVG Preview**
  - **vscode-pdf**
  - **filesize**
    - Shows file size info + other metadata interface
- Copilot
  - **GitHub Copilot**
  - **GitHub Copilot Chat**

Miscellaneous Tips

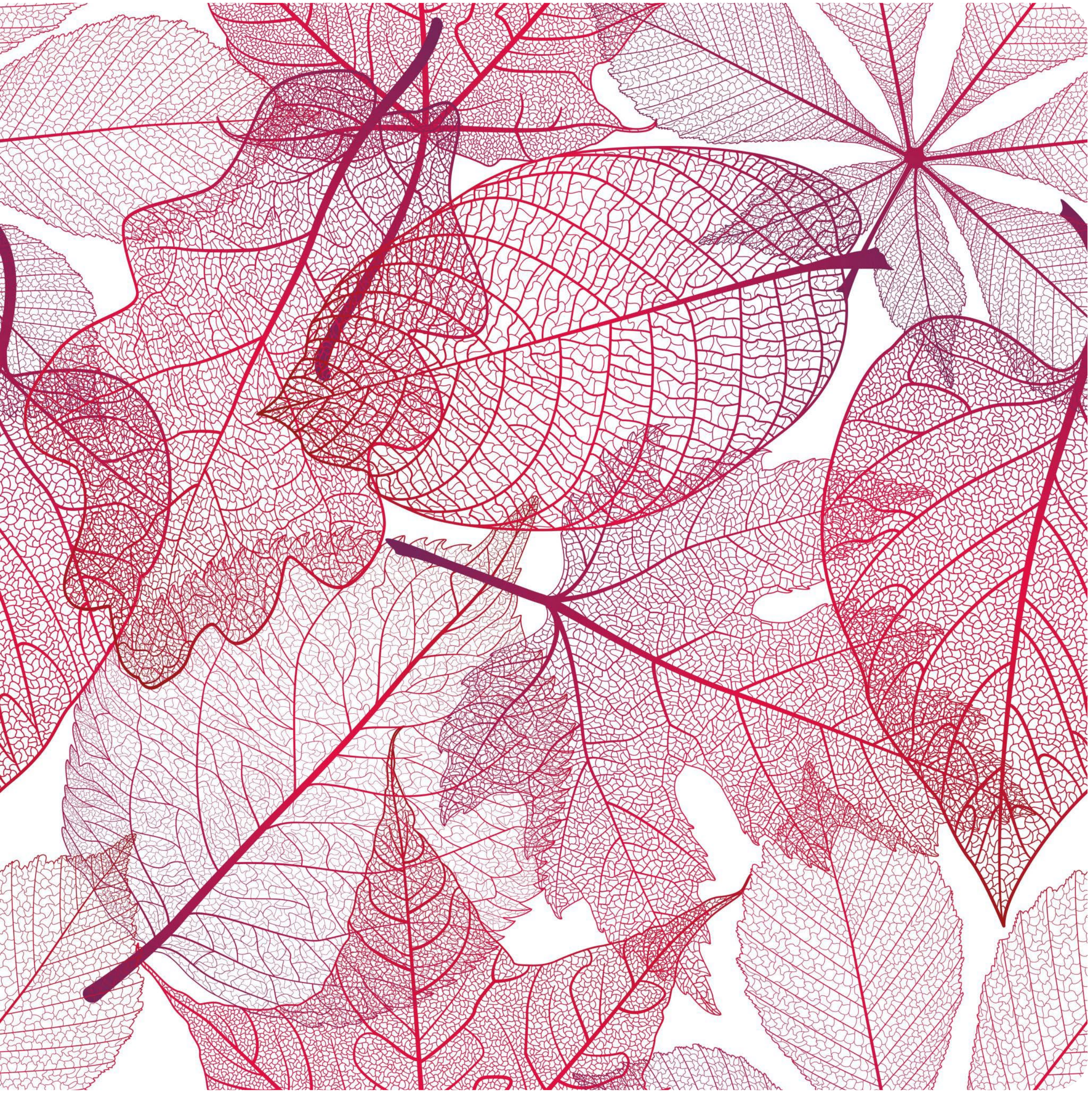
- Get one of the Duke BioRender subscriptions for quick figures
  - Bio/EvAnth, Chemistry
  - NIH BioArt – free alternative
- Get the free Adobe suite from Duke OIT
- Connect GitHub Copilot to VSCode
- Add your ssh key to Duke IT to avoid entering a password every time





Takeaways from my research journey (so far)

- Anyone can learn to code!
- Seek out good mentors in broad fields
- Data wrangling and Unix shell scripting helps you do science reproducibly and efficiently
- Data visualization helps you understand your science and make discoveries
- Enjoy the ride!



Thank you!

kayla.wilhoit@duke.edu